SILENT, UNSEEN GUARDIANS OF THE SEAS

A comparative look at the submarine capabilities of India, China and Pakistan



Beijing's ambitions to project power in the Indo-Pacific region. The country has also been deploying its conventional and nuclear submarines in the Indian Ocean Region (IOR), controlling which is increasingly deemed important for India's geostrategic aims. And naval experts believe the navy must have a full fleet of submarines to project India's power in its backyard.

But China is not India's only challenge. In 2015, Pakistan signed a contract with China to build eight Hangor-class submarines, an export variant of the Yuan-class submarines. In April 2024, China completed the first one. Pakistan aims to acquire all eight submarines by 2028, which could challenge India in terms of underwater warfare capability. Experts believe the stealth submarines for Pakistan are Beijing's way of "building a pressure point" on India. With AIP technology and advanced sensors, the Hangor-class subs have a clear advantage over India's Scorpene-class vessels—the most advanced conventional subs the Navy possesses. This is reason enough for the government to expedite the P75(I) project. Vice Admiral A.K. Singh (retd), who commanded India's first nuclear-powered submarine INS *Chakra*, says unequivocally, "Our underwater capabilities are pathetic; it's high time we speed up."

JOURNEY OF INDIAN SUBMARINES

Throughout the 1970s, India expanded its submarine fleet with Foxtrot-class submarines like the *Kalvari*. From 1986, the Navy took a significant step forward with the acquisition of the Shishumar-class submarines from West Germany. These Type 209 submarines were more advanced, featuring greater endurance, better sensors, and improved weaponry.

In the '80s and '90s, the navy acquired the Sindhughosh-class (Kilo-class) submarines from the erstwhile Soviet Union and Russia. These submarines, equipped with advanced sonar systems and the capability to launch cruise missiles, significantly enhanced India's offensive capabilities. Once the backbone of India's conventional submarine fleet, they are still in service, albeit with upgrades.

The early 2000s marked a new era in India's submarine development, with the pursuit of nuclear-powered submarines capable of launching ballistic missiles. This ambition was realised with the commissioning of the SSBN INS *Arihant* in 2016. Developed under the secretive Advanced Technology Vessel (ATV) project, *Arihant* made India only the sixth nation to possess a nuclear-powered ballistic missile submarine, a key element of the country's nuclear triad—the capability to launch nuclear missiles from land, air and sea. INS *Arighat* is India's second SSBN, and a third, INS *Aridaman*, is in the pipeline.

The Modi government in 2015 ap-

INDIA 16



Nuclear-powered Ballistic Missile Submarines (SSBN)

Numbers: **2 (INS *Arihant*; INS *Arighat*), with INS *Aridaman* under construction**

First inducted in 2016, these can deliver inter-continental ballistic missiles. After the lease, and return, of INS *Chakra* to Russia, India currently has no **Nuclear Attack Submarine (SSN)**, but plans to build two

Diesel-Electric Submarines (SSK) Kalvari-class (Scorpène-class)

Numbers: **5**

Built with France, and inducted in 2017, there are plans to build three more advanced Kalvari-class subs. Fitted with Exocet anti-ship missiles, MICA anti-air missiles

Shishumar-class

Numbers: **4**

German-origin subs acquired from 1986-1994, have good endurance and sensors, Harpoon anti-ship cruise missiles

Sindhughosh-class (Kilo-class)

Numbers: **7**

These Russian-origin submarines were inducted from 1986. They have advanced sonar, can launch anti-ship cruise missiles

CHINA 50+

Jin-Class (SSBN)

Numbers: **6**

First commissioned in 2007, these are a crucial part of China's nuclear deterrence strategy.

These SSBNs can carry up to 12 JL-2 ballistic missiles of 7,000 km range. The under



Total number



"We need to beef up our underwater capabilities. And unless we find a third submarine building shipyard, Project 75(I) won't be a reality"

VICE ADMIRAL A. K. SINGH
Former commander, INS *Chakra*